

Uncertainty-Aware Multimodal Anomaly Detection for Microservice Systems With Active Learning

Hui Dou , Shenglai Guo, Lele Zou, Yiwen Zhang , Pengfei Chen , and Zibin Zheng , *Fellow, IEEE*

Abstract—Accurate and robust anomaly detection is critical for microservice system reliability. Recent multimodal approaches have improved detection comprehensiveness by integrating metrics, logs, and traces. However, they often overlook intra-modal uncertainty from noise, ambiguity, or missing data, and inter-modal uncertainty arising from varying predictive capabilities across modalities. Additionally, extensive labeling of multimodal data remains costly. To address these limitations, we propose MUAD, an uncertainty-aware multimodal anomaly detection framework with active learning. MUAD employs a Graph-based Probabilistic Encoder (GPE) to model intra-modal uncertainty through probabilistic representations, and a Confidence-aware Fusion Mechanism (CFM) to dynamically weight modalities based on their prediction confidence. Furthermore, an active learning paradigm iteratively refines the model using high-confidence pseudo-labels and informative samples, maintaining performance under label-deficient conditions. Experiments on three benchmark datasets demonstrate MUAD achieves 98.10% average F1-score, outperforming state-of-the-art methods by up to 7.87%. Results also confirm its robustness under low-quality data and limited labels.

Index Terms—Microservice systems, anomaly detection, multimodal data, active learning.

I. INTRODUCTION

IN RECENT years, microservices architecture has gained widespread adoption in modern application development. By decomposing tightly coupled applications into independently deployable services, this approach enhances both flexibility and scalability. However, as system scale and complexity increase, the risk of anomalies escalates correspondingly. Anomalies originating in a single service may propagate through the call chain and adversely affect the performance of dependent services, thereby degrading the user experience and potentially leading

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	 detected unusual patterns	 no detected unusual patterns	
System Fault Type			
Resource Underprovisioning			
Database Query Failure			
Network Congestion			

Fig. 1. Abnormal patterns in multimodal data for different system failures. This figure illustrates how different fault types manifest across metrics, logs, and traces. For instance, resource underprovisioning manifests primarily in metrics, while database query failures are evident in logs and traces.

to economic losses [1]. Consequently, timely detection and remediation of performance anomalies is essential for maintaining system reliability and availability.

The observability data collected by operators in microservice systems, encompassing metrics, logs, and traces, plays a pivotal role in ensuring system reliability and security. Specifically, logs record detailed information during service operation, including requests, errors, exceptions, and other events. Metrics, on the other hand, provide a quantitative representation of the system's performance and health status, with common examples including response time, error rate, and CPU utilization. Trace data captures the complete path of a request as it usually flows through multiple microservices. By utilizing trace IDs and span IDs, it reveals invocation relationships and provides a global view of service interactions, aiding in the analysis of service dependencies and performance bottlenecks. Over these years, many studies have successfully utilized unimodal data (i.e., solely metrics [2], [3], logs [4], [5] or traces [6], [7]) to achieve automated anomaly detection, demonstrating certain effectiveness in specific detection tasks. However, due to the limited information provided by each individual modality, unimodal approaches exhibit significant limitations in identifying particular types of anomalies [8], [9], [10]. For example, log data may sometimes reflect minor errors that have little impact on the overall system performance, which can easily trigger false alarms and lead to false positives. Fig. 1 provides an illustrative example of how different modal data may manifest under various anomaly scenarios in microservice systems. Certain business-level anomalies, such

as database query exceptions or code defects, often do not cause significant fluctuations in metrics data, yet they are more readily identifiable through logs and trace data. In contrast, network congestion anomalies are typically evident in both metrics and trace data. Therefore, analytical methods that comprehensively leverage multimodal data can compensate for the shortcomings of single data sources, thereby ensuring the comprehensiveness of anomaly detection.

Given the limitations of unimodal anomaly detection approaches, several studies have proposed multimodal anomaly detection methods in recent years [11], [8], [12], [13]. Although these approaches obtained prospective performance, they still suffer from the following challenges:

1) *Intra-modal uncertainty*: Multimodal data originates from heterogeneous sources and is prone to data uncertainty due to modality noise, absence, and intrinsic ambiguity. These uncertainties directly compromise data quality, thereby diminishing the model's ability to comprehend multimodal content effectively. Existing approaches [11], [8] typically employ deterministic point embeddings to learn modality-specific representations. However, such methods rely heavily on clean or high-quality data. When processing low-quality data, uncertain sample embeddings may cause the model to inaccurately capture critical information, consequently limiting both performance and robustness.

2) *inter-modal uncertainty*: Although each modal data deserves to be paid enough attention in order to detect various anomalous patterns in practical scenarios, the confidence level of each modality is not static across different samples, which are termed as "inter-modal uncertainty" in our paper. Existing multimodal anomaly detection [11], [14], [15] methods inherently assume that all modalities contribute positively to the detection process. However, when confronted with inter-modal conflicts, these approaches prevent the dynamic assessment of which modality can most accurately reflect the system state, thereby lacking the ability to dynamically assess and adjust the confidence levels of each modality.

3) *Paucity of training data*: Fully labeled high-quality samples can significantly enhance the detection performance of models but require extensive manual annotation, which is both time-consuming and costly. Particularly in the context of multimodal data, the complexity and workload increase significantly due to the need to integrate information from heterogeneous sources and make comprehensive judgments about the system state. Consequently, a key challenge in multimodal anomaly detection lies in balancing the trade-off between annotation cost and detection performance, so as to maintain effectiveness under label-deficient conditions.

In this work, we propose MUAD, an uncertainty-aware multimodal anomaly detection framework enhanced with active learning. It consists of four stages: (1) **Unimodal Feature Encoding**, where each modality is encoded into deterministic point-wise features; (2) **Uncertainty Modeling**, which derives robust modality-specific representations by sampling stochastic latent vectors; (3) **Dynamic Modality Fusion**, where a Confidence-aware Fusion Mechanism (CFM) adaptively assigns weights to each modality based on confidence estimates and fuses them into

a global representation; (4) **Active Training**, which iteratively updates the model using high-confidence pseudo-labeled data and highly informative manually labeled samples until the query budget is exhausted. Specifically, to address the first challenge, we design a Graph-based Probabilistic Encoder (GPE) module to extract robust feature representations from complex multimodal content. For the second challenge, CFM enables the model to dynamically focus on the most discriminative modalities. For the third challenge, we incorporate an active learning framework for iterative model training that maintains high detection performance even under limited labeling resources.

We conducted extensive experiments on two widely adopted benchmark microservice platforms (TrainTicket [16] and Social-Network [15]) and one public dataset (SN from Eadro [11]). The experimental results demonstrate that our approach achieves an average F1-score of 98.10%, surpassing previous state-of-the-art unimodal and multimodal methods by up to 7.87%. These results also highlight the high robustness and resilience of MUAD, as it maintains excellent detection performance even under conditions of low-quality modal data and limited labeled samples.

In summary, our work makes the following contributions:

- We propose MUAD, a novel multimodal anomaly detection framework for microservice systems that leverages metrics, logs, and traces. By jointly modeling intra-modal and inter-modal uncertainties, MUAD significantly improves the accuracy and robustness of anomaly detection.
- MUAD employs a graph-based probabilistic encoder to robustly learn information representations for each modality, and dynamically integrates these modalities through a confidence-aware fusion mechanism, achieving efficient multimodal data integration.
- We innovatively integrate active learning into the multimodal anomaly detection setting, which enables MUAD to maintain strong detection performance even under label-deficient conditions. This is particularly important for multimodal anomaly detection, as the complexity of multimodal data typically incurs high annotation costs.
- We conduct extensive experiments on three benchmark datasets, where MUAD achieves state-of-the-art performance. The source codes and datasets utilized in MUAD is open-sourced as a GitHub project [17].

The rest of this paper is organized as follows. Section II introduces the key motivations. Section III describes the detail design and implementation of MUAD. After that, Section IV describes the experimental setups and presents the experimental results and analysis. Section V introduces the related work. Finally, Section VI concludes this paper.

II. MOTIVATION

Motivation #1. It is necessary to model intra-modal data uncertainty to enhance the robustness of multimodal anomaly detection: In real-world scenarios, the collected data is subject to various uncertainties arising from environmental disturbances, limitations in the performance of monitoring components, and human errors. As shown in Fig. 2, intra-modal data uncertainty primarily includes modality noise, absence, and intrinsic

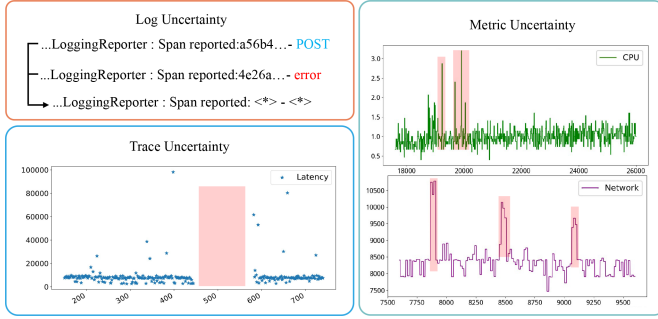


Fig. 2. Examples of low-quality multimodal data in microservices. Log Uncertainty (Left Top): the consistent log templates used for extracting both normal and abnormal events result in ambiguity at the event level. Metric Uncertainty (Right): unusual fluctuations in CPU and network metrics occur during normal system operation, introducing noise that may not indicate actual anomalies. Trace Uncertainty (Left Bottom): missing latency data in call chains compromises the integrity and completeness of trace information.

ambiguity. Specifically, since normal and anomalous logs may share the same log events, this event-level ambiguity makes it difficult for existing log event-based feature extraction methods to effectively distinguish between normal and anomalous patterns, thereby increasing the risk of false positives and false negatives. Moreover, we observed that the presence of noisy metric data further confounds the identification of genuine anomalies, as transient fluctuations induced by factors such as resource contention do not necessarily indicate system anomalies and should not trigger alerts. Additionally, real-world microservice systems often face issues with missing multimodal data, which not only undermines the overall integrity of the data but also significantly impacts the quality of feature representation. In particular, the absence of call chain data adversely affects the learning of modal graph features that rely on call chain generation. These findings suggest that intra-modal data uncertainty diminishes the model's ability to comprehend multimodal content, thereby necessitating a robust feature extraction method to effectively model such uncertainty.

Motivation #2. It is necessary to deal with inter-modal uncertainty to ensure the reliability of multimodal anomaly detection: Due to the various anomalous patterns, the confidence level of each modality in anomaly detection is inherently non-stationary across different samples. To validate this conclusion, we utilize the trained MUAD model to predict the True Classification Probability (TCP) values for each modality across two different fault sample sets and two different normal sample sets from Dataset $\mathcal{B}$. Specifically, TCP is derived from the softmax output of the modality-specific classifier, representing how confidently each modality predicts the correct system state. Higher TCP values (closer to 1.0) indicate more reliable modality predictions (Detailed formulation can be found in Section III-D). Besides, Dataset $\mathcal{B}$ was collected locally from the open-source microservice systems SocialNetwork [15] and detailed can be found in Section IV-A. We focused on two fault types: CPU exhaustion and Database query failure, as well as another two separate sample sets from the normal data. These experimental setups enable us to evaluate the confidence levels of different modalities under various conditions. As shown in Fig. 3, the

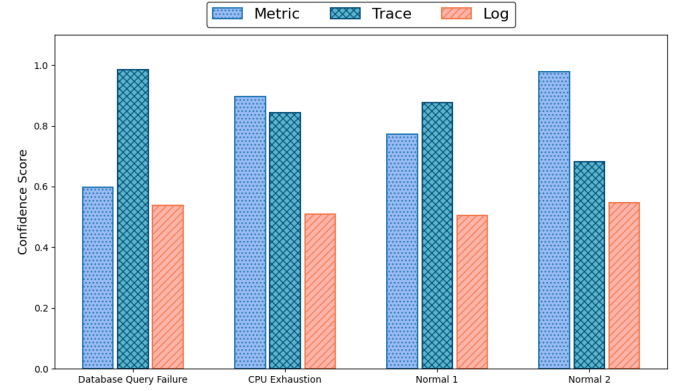


Fig. 3. Based on the experimental results of MUAD on Dataset $\mathcal{B}$, we can find that the confidence scores of different modalities vary dynamically across samples. Data from low-confidence modalities may constrain optimal fusion strategies, thereby adversely affecting system state inference.

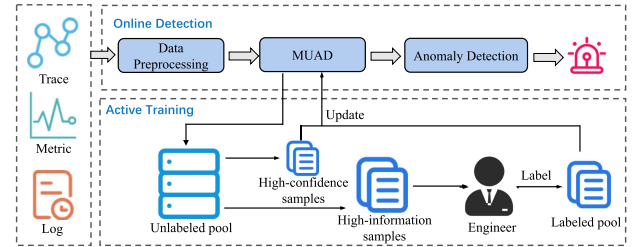


Fig. 4. The overview framework of MUAD. The framework operates in an iterative training paradigm with active learning. In each iteration, multimodal data (metrics, logs, traces) are processed by MUAD for anomaly detection. The labeled pool is updated through two mechanisms: high-confidence samples are automatically pseudo-labeled, while high-informative samples are manually annotated by engineers. This process continues until the query budget is exhausted.

metrics demonstrate high confidence scores in detecting CPU overload conditions, indicating their effectiveness in inferring system status. However, they become unreliable when diagnosing database query anomalies, where traces obtain the highest confidence score. If proper weight allocation is not implemented during multimodal fusion, these metrics may dominate the overall detection decision, posing a risk to detection robustness. In such cases, greater emphasis should be placed on modalities with higher prediction confidence to ensure the reliability of anomaly detection results. Notably, logs consistently exhibit low confidence scores, primarily because log-based anomaly detection methods often fail to meet the precision requirements of practical applications [8], [14]. These critical observations motivate our introduction of a dynamic fusion mechanism into the multimodal anomaly detection framework, enabling automatic identification of the most trustworthy information sources and consequently enhancing overall detection accuracy.

III. APPROACH

A. Design Overview

Fig. 4 presents an overview of the MUAD framework, a multimodal anomaly detector integrated with an active learning strategy. The framework operates in an iterative training paradigm, wherein human annotators label the most informative

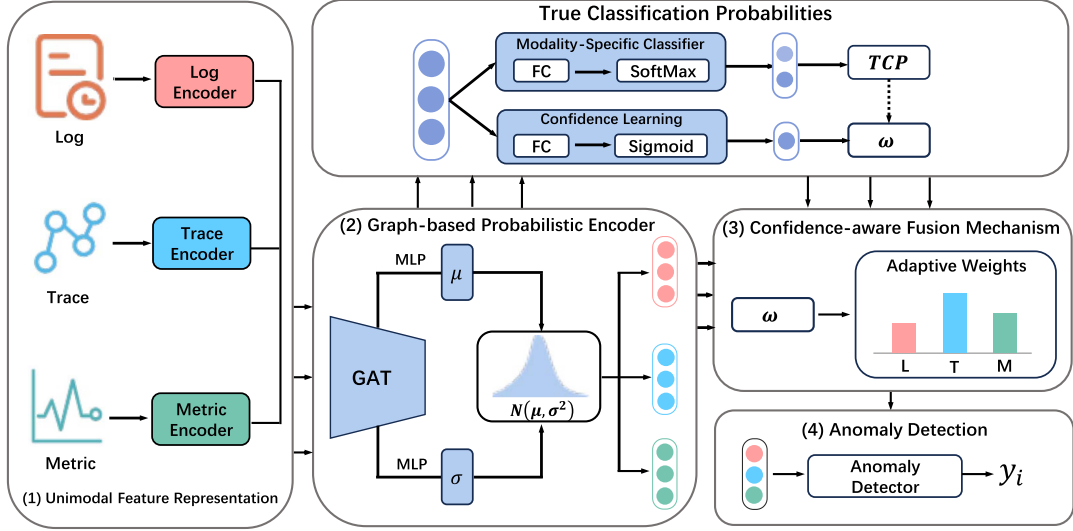


Fig. 5. Major stages in MUAD for anomaly detection by modeling both intra-modal and inter-modal uncertainties. The pipeline consists of four stages: (1) Unimodal Feature Representation extracts modality-specific features using dedicated encoders; (2) Graph-based Probabilistic Encoder (GPE) models intra-modal uncertainty by mapping features to Gaussian distributions $N(\mu, \sigma^2)$ through GAT and MLP layers; (3) Confidence-aware Fusion Mechanism (CFM) dynamically assigns fusion weights based on confidence scores, thereby modeling inter-modal uncertainty; (4) Anomaly Detector produces the final prediction.

samples and high-confidence predictions are automatically assigned pseudo-labels to expand the annotated dataset. Due to the innovative introduction of confidence prediction into the active learning process, MUAD can significantly reduce the effort of manually labeling data. Within a given query budget, the model aims to maximize detection performance. During each iteration, MUAD adopts the approach illustrated in Fig. 5 to jointly model both intra-modal and inter-modal uncertainties, thereby ensuring the detector's performance and robustness. Specifically, modality-specific encoders are first applied to process inputs from different data sources. To capture intra-modal uncertainty, a Graph-based Probabilistic Encoder (GPE) is employed to represent input features as Gaussian distributions. Subsequently, a Confidence-aware Fusion Mechanism (CFM) is designed to adaptively assign fusion weights to each modality. The final fused representation is then forwarded to the anomaly detector for prediction.

B. Unimodal Feature Representation

In microservice systems, different observable data typically exist in heterogeneous forms, including logs, metrics, and traces. To capture modality-specific representations, we design individual unimodal feature extractors tailored to each modality.

Metric Feature Representation: The metrics collected from each microservice are presented as a multivariate time series, reflecting the performance status of the corresponding service. To efficiently capture the key dynamic information, we employ a Gated Recurrent Unit (GRU) to model the intramodality dynamics in the time series, using its final hidden state to represent the entire sequence:

$$f_m = GRU(\{M_1^k, M_2^k, \dots, M_T^k\}), f_m \in \mathbb{R}^{D^t} \quad (1)$$

where $\{M_1^k, M_2^k, \dots, M_T^k\}$ denote the multivariate time series composed of k metrics over an observation window of size T .

f_m represents the metric feature, and D^t denotes the output dimension of the feature extractor.

Log Feature Representation: Following the log parsing approach described in previous work, we first use Drain, a widely adopted log parsing tool in AIOps research community, to convert unstructured log messages into structured log events. Subsequently, we model the occurrence of these log events using the Hawkes process [18], as it has demonstrated both effectiveness and efficiency in prior evaluations of log event representation learning [11]. To ensure that the resulting log features are compatible with subsequent processing stages, we project them into a unified D -dimensional space using a fully connected layer with a ReLU activation function. Finally, we extract deterministic log embeddings, denoted by $f_l \in \mathbb{R}^{D^t}$.

Trace Feature Representation: We follow up-to-date previous works [11], [19] and capture delay times to represent traces. Specifically, we process all traces within each sliding window to extract the average delay associated with each service and construct the corresponding time series data. Similarly, a GRU is employed to generate the trace feature representation, denoted by $f_t \in \mathbb{R}^{D^t}$.

C. Graph-Based Probabilistic Encoder

Existing multimodal representation methods encode each modality as a deterministic point in the embedding space, thereby overlooking the uncertainty inherent in the data. To address this limitation, we model the input features of each modality as Gaussian distributions and reconstruct their embedding representations through distribution sampling. Notably, given the characteristics of anomaly propagation among microservices, we focus on capturing graph-level features for each modality to sense the topological structure over a broader neighborhood, rather than reconstructing independent embeddings for each service. Specifically, we employ a Graph Attention

Network (GAT) to learn the salient information of each modality within the graph structure and utilize a pooling layer to obtain the graph-level features, i.e., $\{g_m, g_l, g_t\}$. However, these features remain deterministic and fail to capture the uncertainty present in multimodal data. Our GPE module further models the input features of each modality as Gaussian distributions, which can be formally expressed as:

$$\mu_v, \sigma_v^2 = MLP(g_v), v \in \{m, l, t\} \quad (2)$$

$$p(z_v|x) = \mathcal{N}(\mu_v, \sigma_v^2) \quad (3)$$

where the mean vector (μ) represents the feature representation identified by each modality, and the variance vector (σ) indicates the range of the modality's distribution over each dimension—the larger the variance, the higher the uncertainty of the observed content. Now, instead of a deterministic point embedding, we can sample a random embedding z_v from $p(z_v|x)$. However, since the sampling operation is non-differentiable, we employ the reparameterization trick to allow gradient descent to back-propagate normally during parameter optimization. Specifically, for each modality, we first sample random noise ξ from a standard normal distribution and then generate z_v as an equivalent sampled representation:

$$z_v = \mu_v + \xi \sigma_v, \xi \sim \mathcal{N}(0, I) \quad (4)$$

Although GPE is capable of estimating the mean and variance for each modality, if the variance becomes too small during training, the modality feature representations can degenerate into deterministic embeddings. To counteract this effect, we introduce a KL divergence regularization term to constrain each modality's distribution to approach the standard normal distribution, $\mathcal{N}(\mathbf{0}, I)$. Specifically, the optimization is formulated as follows:

$$\begin{aligned} \mathcal{L}_{kl} &= \frac{1}{3} \sum_v KL(\mathcal{N}(\mu_v, \sigma_v^2) || \mathcal{N}(\mathbf{0}, I)) \\ &= \frac{1}{3} \sum_v \frac{1}{2} (\mu_v^2 + \sigma_v^2 - \log \sigma_v^2 - 1) \end{aligned} \quad (5)$$

where $KL(\cdot)$ denotes the divergence between two probability distributions. Under the guidance of this loss function, the model is discouraged from adopting excessively low variances, thus preserving the uncertainty in the multimodal features and preventing their collapse into deterministic representations.

D. Confidence-Aware Fusion Mechanism

As the varying importance of different modalities in predicting the final system state, a dynamic mechanism is required to weigh their contributions during the fusion process. Here we adopt a confidence-aware fusion mechanism (CFM) that adaptively estimates modality-specific weights. As shown in Fig. 5, CFM applies a modality-specific classifier to each modality after obtaining its uncertainty representation. This classifier generates a softmax output, and the module selects the true classification probability (TCP) [20], [21] from the softmax outputs as the confidence score. Unlike using the maximum softmax probability, which may lead to overconfidence when

a modality makes an incorrect prediction, TCP uses the softmax probability corresponding to the actual label, thereby mitigating such overconfidence. Formally, for each modality, the confidence score can be expressed as:

$$TCP^v = y \cdot p(y | z_v) = \sum_{k=1}^K y_k p_k^v \quad (6)$$

where $(\cdot)$ denotes the inner product operation, and y_k represents the k element of the label. It can be observed that $TCP^v \in (0, 1)$ reflects the correctness of a modality's information in representing the system state. A higher value of TCP^v indicates that the modality's features accurately capture the system state, whereas a lower value suggests incorrect predictions. However, ground-truth labels are unavailable during the testing phase, making it infeasible to use TCP^v directly as a weight. To ensure consistency between training and testing, a confidence network is constructed to predict confidence scores that will serve as the modality fusion weights:

$$\omega_v = Sigmoid(MLP(z_v)) \quad (7)$$

The confidence network is implemented as a lightweight architecture: a single fully-connected layer that maps $z_v \in \mathbb{R}^D$ to a scalar, followed by a Sigmoid activation to ensure $\omega_v \in (0, 1)$. This simple design prevents overfitting while enabling efficient confidence estimation during both training and inference.

To achieve the above objective, we employ the following loss function:

$$\mathcal{L}_p^v = - \sum_{i=1}^n \log(g_i^v(I_i^*)) \quad (8)$$

$$\mathcal{L}_q^v = \sum_{i=1}^n MSE(TCP_i^v, \omega_i^v) \quad (9)$$

$$\mathcal{L}_{con} = \sum_{v \in \{m, l, t\}} (\mathcal{L}_p^v + \mathcal{L}_q^v) \quad (10)$$

In practice, $\mathcal{L}_p^v$ and $\mathcal{L}_q^v$ are jointly optimized through end-to-end training. During each forward pass, the classifier g^v generates softmax outputs, from which TCP^v is computed and used as the supervision signal for training the confidence network. Both the classifier and confidence network are updated simultaneously via backpropagation through (10), where the two loss components play complementary roles: $\mathcal{L}_p^v$ (8) ensures accurate classification through cross-entropy loss on the ground-truth label I_i^* , while $\mathcal{L}_q^v$ (9) trains the confidence predictor ω^v to estimate the classifier's reliability via mean squared error with TCP^v . This joint optimization enables each modality to develop reliable self-assessment capabilities for optimal multi-modal fusion.

Finally, we aggregate the representations from all modalities into a unified system state representation by performing a confidence-weighted fusion: $h = \sum \omega_v z_v$.

E. Anomaly Detection

Finally, the anomaly detector uses the fused representation as input to to classify the current system state as either normal or

anomalous. We formulate this explicitly as a binary classification task. Formally, this can be expressed as:

$$\hat{y} = \operatorname{argmax}[\operatorname{Softmax}(Wh + b)] \quad (11)$$

In the equation above, W and b respectively denote the trainable parameters, and $\hat{y} \in \{0, 1\}$ represents the predicted discrete label, where 0 denotes a normal state and 1 denotes an anomaly. If an anomaly is detected, MUAD triggers an alert and provides the modality confidence scores to the operator, facilitating the assessment of the alert's reliability.

F. Active Training

To balance labeling cost and model performance, we apply active learning to train our model. As a label-efficient learning paradigm, active learning selects the most informative samples (MIS) for annotation under a given query budget, aiming to maximize model performance at minimal labeling cost. Most existing studies on active anomaly detection focus on selecting samples for expert annotation, while overlooking the value of high-confidence samples (HCS) in accelerating model training. However, if the model exhibits sufficient confidence in its predictions, pseudo-labels can be automatically assigned and incorporated into the training process.

The active training procedure of MUAD is described in Algorithm 1. To fully leverage the above advantages, we design a novel active learning query strategy hybridly utilizing the MIS and HCS. Specifically, high-informative samples are identified by calculating the prediction entropy:

$$H(x_i) = - \sum_c p(c | x_i) \log p(c | x_i) \quad (12)$$

where higher entropy indicates greater model uncertainty. These samples, termed as most informative samples (MIS) in our framework, are selected for expert annotation. For high-confidence samples (HCS), the predicted confidence score $\widehat{TCP}_i$ (defined in Eq. (6)) serves as the confidence indicator, and samples with $\widehat{TCP}_i > \theta$ are automatically assigned pseudo-labels via $\arg \max p(y|x_i)$. These two mechanisms work complementarily to balance annotation cost and detection performance controlled by the confidence threshold θ . Based on the above query strategy, the active training process proceeds as follows. We begin by randomly labeling 10% of the data to initialize the MUAD model. During each subsequent iteration, we select MIS by computing the entropy of the model's predictions. High entropy typically indicates that the model is uncertain about the sample's classification, highlighting areas where the current model struggles and where expert labeling is most valuable. Next, we perform confidence prediction and label inference on the remaining unlabeled data, and incorporate samples with confidence scores exceeding a predefined threshold θ (i.e., HCS) into the training set for the next iteration. Since the model is essentially a classifier, we adopt the $\widehat{TCP}$ value as an indicator of the model's confidence in its predictions. In each learning phase, the model's training loss consists of the following components:

$$\mathcal{L}_{MUAD} = \mathcal{L}_{ano} + \lambda_1 \mathcal{L}_{kl} + \lambda_2 \mathcal{L}_{conf} + \mathcal{L}_{mse} \quad (13)$$

Algorithm 1: Active Training Procedure of MUAD.

Input: Unlabeled pool $\mathcal{N}_U$; Labeled pool $\mathcal{N}_L$; High-confidence sample set; $\mathcal{N}_C$ Query budget b ; Query batch size k ; Confidence threshold θ .
Output: Anomaly detection model MUAD and Labeled dataset $\mathcal{N}_L$.

```

1  $\mathcal{N}_C = \emptyset$ ;
2 while  $|\mathcal{N}_U| < b$  do
3    $\triangleright$  Stage 1: Training the MUAD Model
4   begin
5      $\mathcal{N}_L \leftarrow \mathcal{N}_L \cup \mathcal{N}_C$ ;
6     Train the anomaly detection model  $\triangleright$ Eq. (13);
7   end
8
9    $\triangleright$  Stage 2: Labeling High-Informative Samples
10  begin
11    Calculate the entropy values in  $\mathcal{N}_U$ ;
12    Select the Top-k samples for annotation;
13    Update  $\mathcal{N}_L$ ;
14  end
15
16   $\triangleright$  Stage 3: Selecting High-Confidence Samples
17  for  $x_i \in \mathcal{N}_U$  do
18    if  $\widehat{TCP}_i > \theta$  then
19      Assign pseudo-labels;
20      Update  $\mathcal{N}_C \leftarrow x_i$ ;
21    end
22  end
23   $\mathcal{N}_L \leftarrow \mathcal{N}_L \cup \mathcal{S}$ ,  $\mathcal{N}_C \leftarrow \mathcal{N}_C \cup \mathcal{H}$ ,  $\mathcal{N}_U \leftarrow \mathcal{N}_U \setminus \mathcal{S}$ ;
24 end
25 return  $\mathcal{N}_L$ 

```

where $\mathcal{L}_{ano}$ denotes the binary cross-entropy loss on labeled data, and $\mathcal{L}_{mse}$ represents the loss for label confidence estimation. λ_1 and λ_2 are the weighting coefficients for the corresponding loss components. We repeat the above iterative process until the query budget b is exhausted.

IV. EXPERIMENT

In this section, we evaluate MUAD by answering the following research questions (RQs) through empirical studies:

- *RQ1*: How effective is MUAD in anomaly detection tasks?
- *RQ2*: How does each component contribute to MUAD's overall performance?
- *RQ3*: Does MUAD exhibit high robustness?
- *RQ4*: Is it necessary to consider high confidence samples in MUAD's active learning strategy?
- *RQ5*: How do MUAD's hyperparameters affect its performance?

A. Experimental Setup

1) *Datasets*: To evaluate the performance of MUAD, we conduct extensive experiments on three datasets ($\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$), as summarized in Table I. We randomly selected 60% of the

TABLE I
DATASET COMPOSITION

Dataset	Metric Length	Log Messages	Trace Quantity
Dataset $\mathcal{A}$	473575	386707	11132
Dataset $\mathcal{B}$	290184	216249	33545
Dataset $\mathcal{C}$	16944	15365	11345

data from each of the three datasets for training and used the remaining 40% for testing, ensuring that the ratio of normal to abnormal samples remains consistent between the training and testing sets.

Dataset $\mathcal{A}$ and Dataset $\mathcal{B}$ were collected locally from the opensource microservice systems TrainTicket [16] and SocialNetwork [15], respectively. TrainTicket is a platform that provides railway ticketing services, while SocialNetwork simulates a microservice-based social networking application. Both have been widely studied in prior research on microservice anomaly detection. Following previous studies [11], [22], we adopt chaos engineering techniques to inject seven types of anomalies: CPU/memory/disk stress, service failure, database query exceptions, packet loss, and packet loss/delay. During our data collection phase, each anomaly lasts for two minutes. In total, we perform 133 and 84 injection operations on Dataset $\mathcal{A}$ and Dataset $\mathcal{B}$, respectively. Metrics, logs, and trace data are collected through three separate collectors.

Dataset $\mathcal{C}$ is the *SN* dataset from Eadro [11], which provides a multimodal dataset suitable for microservice anomaly detection. It contains various faults, such as CPU overload, network packet loss, and network latency. The dataset includes 7 metrics sampled every 1 s, 126,384 traces, and 15 log templates.

2) *Baselines*: We compare MUAD against representative state-of-the-art methods in microservice anomaly detection, including both unimodal and multimodal approaches. These baselines were selected based on their technical diversity, recency, and reproducibility, ensuring comprehensive evaluation across different modality combinations and detection paradigms. The selected baselines are as follows:

- MTAD-GAT [23] is a multivariate time series anomaly detection method that leverages graph attention networks to capture temporal and inter-variable dependencies of metrics.
- HADES [8] is a semi-supervised multimodal anomaly detection model that takes logs and metrics as inputs and fuses modality features via a cross-modal attention mechanism.
- MRCA [19] performs automated fault localization using multimodal data, mainly logs and traces, and detects anomalous services based on reconstruction probabilities.
- UAC-AD [24] is an unsupervised adversarial contrastive learning-based method that enhances anomaly detection by distinguishing hard and abnormal samples.
- MSTGAD [12] proposes a multimodal anomaly detection framework that integrates metrics, logs, and traces through attentive multi-modal learning.
- Eadro [11] is a supervised multimodal anomaly detection approach that nonlinearly fuses features from three

modalities and employs a graph attention network to model dependency-aware system states in microservices.

- DeAnomaly [25] is a two-phase unsupervised anomaly detection framework that decomposes multivariate time series to remove normal patterns and uses diffusion models on residuals for robust and accurate anomaly detection.
- HG-PAD [26] is an unsupervised framework that models microservice dependencies via heterogeneous graphs, employing GNN, CVAE, and GAT for anomaly detection using metrics and traces. We implement this method following the original paper as no official code is available.
- Medicine [10] is a multimodal failure diagnosis approach that mitigates modality dominance through independent encoding and adaptive optimization, achieving robust performance under missing or low-quality modality data.

3) *Performance Metrics*: Since anomaly detection is essentially a binary classification task, we adopt Precision, Recall, and F1-score to evaluate the performance of MUAD and other baseline models. These metrics are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (14)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (15)$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (16)$$

where TP (True Positives) denotes the number of correctly identified anomalous samples, FP (False Positives) refers to the number of normal samples incorrectly identified as anomalous, and FN (False Negatives) represents the number of anomalous samples that were not detected.

4) *Implementation and Settings*: We implement the prototype of our method using Python 3.8. All experiments are conducted on a Linux server equipped with an NVIDIA GeForce GTX 3060 GPU. The experimental hyperparameters include the sliding window length, stride, confidence threshold θ , and manually tuned loss weights. Specifically, we set the sliding window length to 10 and the stride to 1. It is worth noting that microservice modalities are sometimes asynchronous in practice. While data alignment is necessary for multimodal fusion, the small stride allows MUAD to capture transient anomalies that might otherwise be smoothed out, and the Hawkes process implicitly preserves the asynchronous arrival dynamics of logs within each window. The threshold θ to 0.9. MUAD is trained using the Adam optimizer [27] with an initial learning rate of 0.001, a batch size of 50, and 120 epochs per training stage. We continue the iterative process of sample selection and training until the model performance reaches the level of using the fully labeled dataset. For the baseline methods, we adopt the parameters provided in the corresponding papers and official open-source implementations.

B. RQ1: Overall Performance of MUAD

To evaluate the effectiveness of MUAD in anomaly detection tasks, we compare its overall performance and performance under varying labeled data quantities with several representative anomaly detection models, as shown in Table II. Notably, to

TABLE II
PERFORMANCE COMPARISON FOR ANOMALY DETECTION

Approach	Modality			Dataset $\mathcal{A}$			Dataset $\mathcal{B}$			Dataset $\mathcal{C}$		
	Metric	Log	Trace	F1-score	Recall	Precision	F1-score	Recall	Precision	F1-score	Recall	Precision
MTAD-GAT	✓			0.7520	0.8267	0.6897	0.6706	0.6455	0.6978	0.8561	<u>0.9999</u>	0.7484
HADES	✓	✓		0.4275	0.3305	0.6053	0.5631	0.6060	0.5259	0.9084	1.0000	0.8321
UAC-AD	✓	✓		0.4589	0.3606	0.6309	0.5070	0.4236	0.6314	0.4761	0.3255	0.8861
MRCA		✓	✓	0.2090	0.1346	0.4592	0.4355	0.3013	0.7857	0.9412	0.8889	0.9412
MSTGAD	✓	✓	✓	0.7438	0.6994	0.7943	0.8251	0.8261	0.8240	0.9802	0.9920	0.9686
DeAnomaly	✓			0.5259	0.4607	0.6124	0.6788	0.5316	0.9387	0.9753	0.9701	0.9806
HG-PAD	✓		✓	0.4742	0.3477	0.7455	0.5121	0.4004	0.7102	0.8167	0.7154	0.9514
Medicine	✓	✓	✓	0.3663	0.3905	0.4476	0.4143	0.4048	0.4881	0.8056	0.8334	0.8667
Eadro	✓	✓	✓	<u>0.9022</u>	<u>0.8484</u>	<u>0.9632</u>	<u>0.9234</u>	<u>0.8936</u>	<u>0.9551</u>	<u>0.9831</u>	0.9839	<u>0.9823</u>
MUAD	✓	✓	✓	0.9809	0.9789	0.9829	0.9692	0.9593	0.9657	0.9928	0.9941	0.9914
<i>MUAD-10%</i>	✓	✓	✓	0.8447	0.8452	0.8443	0.8729	0.8514	0.8956	0.9314	0.9849	0.8833
<i>MUAD-30%</i>	✓	✓	✓	0.9208	0.9271	0.9145	0.9289	0.9231	0.9348	0.9759	0.9822	0.9697
<i>MUAD-40%</i>	✓	✓	✓	0.9584	0.9625	0.9543	0.9358	0.9421	0.9295	0.9825	0.9844	0.9807

mitigate potential dataset size bias, we selected the closest matching percentage of labeled samples for fair comparison. Based on these results, we make the following observations:

Overall, MUAD outperforms other state-of-the-art anomaly detection methods in terms of F1-score, recall, and precision. Our approach achieves the highest average F1-scores of 0.9809 on Dataset $\mathcal{A}$, 0.9692 on Dataset $\mathcal{B}$, and 0.9928 on Dataset $\mathcal{C}$, surpassing the best-performing baselines by 7.87%, 4.58%, and 0.97%, respectively. Notably, on Dataset $\mathcal{C}$, although MUAD exhibits a slightly lower recall than Hades, it achieves a 24.3% improvement in precision. This suggests that Hades incurs a higher false positive rate, which in turn compromises the overall reliability of the system. These results further confirm MUAD's ability to detect anomalies with minimal false alarms or missed detections, underscoring the effectiveness of the proposed method.

To further understand MUAD's advantages, we analyze the performance of the baseline methods. Single-modality methods like DeAnomaly show strong performance on specific datasets (e.g., F1=0.9753 on Dataset $\mathcal{C}$) but suffer degradation on others (F1=0.5259 and 0.6788 on Datasets $\mathcal{A}$ and $\mathcal{B}$), indicating sensitivity to dataset characteristics and limited generalization. Multimodal methods, while leveraging richer information, do not automatically guarantee superiority. HG-PAD achieves F1-scores ranging from 0.4742 to 0.8167, and Medicine yields 0.3663 to 0.8056 across the three datasets. This suggests that the key challenge lies not in the number of modalities, but in effective fusion under uncertainty. Among all baselines, Eadro achieves the strongest performance (F1=0.9022–0.9831) yet remains constrained by limited uncertainty modeling. In contrast, MUAD explicitly models both intra-modal and inter-modal uncertainties through GPE and CFM, enabling consistently superior performance across all datasets. In addition, MUAD's

representations are robust not only for detection but also for identifying the source of anomalies, paving the way for future end-to-end troubleshooting frameworks. In a preliminary experiment on Dataset $\mathcal{C}$, we extended MUAD with a simple classification head for service-level localization. The extended model achieved a HitRate@1 of 99.24%, outperforming the baseline Eadro (97.59%). Finally, despite the architectural complexity, MUAD remains computationally efficient. For example, it requires only 3.35s for inference on Dataset $\mathcal{C}$ with a consumer-grade NVIDIA RTX 3060 GPU, which is well within the acceptable range for practical anomaly detection tasks.

Due to the active learning strategy specially designed for multimodal anomaly detection model training, MUAD further demonstrates superior performance under label scarcity. As shown in Table II, across all three datasets, our approach is able to achieve an average F1-score of 94.18% using only 30% of labeled data, marginally outperforming the fully supervised Eadro method, which attains 93.62%. Moreover, model performance progressively improves with additional labeled data, confirming MUAD's capability to effectively balance annotation costs and detection performance.

C. RQ2: The Ablation Study of the Proposed Method

To further validate the effectiveness of MUAD, we performed ablation experiments on Datasets $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$. We designed six sets of ablation experiments to evaluate the individual modules of MUAD as follows:

- *Only Metric*: Uses only metric data to evaluate its standalone contribution.
- *Only Trace*: Employs only trace data to assess its impact.
- *Only Log*: Relies solely on log data to measure its effectiveness.

TABLE III
EXPERIMENTAL RESULTS OF THE ABLATION STUDY

Dataset	Approach	F1-score	Recall	Precision
Dataset $\mathcal{A}$	Only Metric	0.9459	0.9306	0.9619
	Only Log	0.6805	0.5916	0.8008
	Only Trace	0.8837	0.8252	0.9511
	w/o GPE	0.9467	0.9367	0.9568
	w/o G	0.9650	0.9607	0.9694
	w/o CFM	0.9651	0.9666	0.9635
	MUAD	0.9809	0.9789	0.9829
Dataset $\mathcal{B}$	Only Metric	0.9452	0.9511	0.9394
	Only Log	0.7350	0.5810	0.7070
	Only Trace	0.8735	0.8455	0.9035
	w/o GPE	0.9484	0.9432	0.9536
	w/o G	0.9524	0.9477	0.9571
	w/o CFM	0.9350	0.9143	0.9086
	MUAD	0.9692	0.9593	0.9657
Dataset $\mathcal{C}$	Only Metric	0.9903	0.9925	0.9882
	Only Log	0.9064	1.000	0.8289
	Only Trace	0.9470	0.9758	0.9199
	w/o GPE	0.9922	0.9919	0.9925
	w/o G	0.9925	0.9962	0.9888
	w/o CFM	0.9909	0.9919	0.9898
	MUAD	0.9928	0.9941	0.9914

- *w/o GPE*: This variant removes the Graph-based Probabilistic Encoder, instead using deterministic point features extracted from each modality.
- *w/o G*: A simplified variant where the graph component is removed from GPE, retaining only the probabilistic encoding for each modality.
- *w/o CFM*: This variant excludes the Confidence-aware Fusion Module, performing anomaly detection by directly concatenating features from all modalities.

As presented in Table III, the superior performance of MUAD can be primarily attributed to the complementary integration of multiple modalities. Relying on a single modality leads to varying degrees of performance degradation across datasets. For Dataset $\mathcal{A}$ and $\mathcal{B}$, MUAD respectively achieves an average F1-score improvement of 14.42% and 11.80%, demonstrating its excellent capability in handling multimodal data. These results further underscore the importance of integrating diverse data modalities to enhance the accuracy of anomaly detection.

Modeling intra-modal uncertainty is crucial for robustly capturing modality-specific features. When the GPE module is removed, the model struggles to extract meaningful information from low-quality multimodal data, leading to degraded overall performance. Moreover, in the experimental results on Datasets $\mathcal{A}$ and $\mathcal{B}$, we observe that w/o G performs worse than w/o GPE, highlighting the benefits of modeling uncertainty in graph-based modality features. Notably, prior studies [28], [29] have also

widely adopted graph-based representations for modality features, further validating the effectiveness of our feature extraction approach.

It is evident that w/o CFM, which relies on simple concatenation rather than the proposed confidence-aware fusion strategy, exhibits varying degrees of performance degradation across different evaluation metrics. This further highlights the superiority of our fusion approach. We attribute this to the fact that different modalities contribute unequally, and the use of a confidence-aware approach enables adaptive learning of weights for each modality, allowing the model to focus more on the most informative modalities for anomaly detection.

Finally, it is worth noting that the ablation results on Dataset $\mathcal{C}$ seems less pronounced, which is mainly due to the dataset's specific characteristics. In detail, the Only Metric variant achieves an F1-score of 0.9903, which is close to MUAD's 0.9928. This is mainly because the smaller size of Dataset $\mathcal{C}$ limits the ability of complex models to fully exploit their structural advantages. Moreover, the fault types in Dataset $\mathcal{C}$ are relatively homogeneous, with resource- and performance-related anomalies (e.g., CPU overload, network latency) accounting for a significant proportion. These anomalies are directly and prominently reflected in performance metrics, enabling the Only Metric variant to achieve strong discriminative capability on its own. As a result, incorporating additional modalities and considering intra-modal and inter-modal uncertainty yields only limited incremental gains for Dataset $\mathcal{C}$.

In summary, the ablation results demonstrate that MUAD's components contribute synergistically to its performance gains. The multimodal integration provides comprehensive observability, while GPE and CFM ensure robust uncertainty modeling and reliable fusion, collectively enabling MUAD to achieve state-of-the-art anomaly detection across diverse scenarios.

D. RQ3: Robustness Evaluation

To comprehensively evaluate the robustness of MUAD under realistic data quality degradation, we select Dataset $\mathcal{A}$, which has the largest data volume among our benchmarks, for experimentation. We examine two data imperfection scenarios prevalent in production microservice systems: **Inter-modal Information Inconsistency** and **Random Modality Missing**. We quantify robustness using relative performance degradation, which normalizes the impact of perturbations against each model's clean baseline:

$$\text{Degradation}(\%) = \frac{F1_{\text{baseline}} - F1_{\text{perturbed}}}{F1_{\text{baseline}}} \times 100 \quad (17)$$

Lower degradation indicates stronger robustness. We compare MUAD against Eadro, the best-performing multimodal baseline from Section IV-B, using this metric alongside absolute F1-scores.

1) *Inter-Modal Information Inconsistency*: In practical microservice systems, multimodal observations may become inconsistent due to heterogeneous noise, asynchronous data collection, or partial modality corruption. To simulate such realistic conditions, we construct an inter-modal information inconsistency setting, where the reliability of different modalities is

TABLE IV
ROBUSTNESS TO INTER-MODAL INFORMATION INCONSISTENCY

Method	Metric	Perturbation Setting		
		Clean	Single-Modal	Multi-Modal
Eadro	F1	0.9022	0.8455	0.7823
	Deg (%)	0.00	6.28	13.29
MUAD	F1	0.9809	0.9584	0.9399
	Deg (%)	0.00	2.29	4.18

unevenly and asynchronously degraded across samples. Specifically, we consider two perturbation scenarios with increasing severity. Perturbations are generated by adding noise factor τ sampled from a standard normal distribution $\mathcal{N}(0, I)$, and scaled according to empirical data statistics, ensuring realistic distortion levels without assuming identical noise patterns across modalities. In the Single-Modality Perturbation setting, we randomly select 30% of the test samples. For each selected sample, we apply the noise injection to a single, randomly chosen modality while keeping others clean. In the Multi-Modality Perturbation setting, we randomly select 40% of the test samples and apply simultaneous noise injection to multiple modalities for each selected sample. By perturbing several modalities at the same time, this setting induces more severe inter-modal inconsistency.

As shown in Table IV, MUAD demonstrates strong robustness under both perturbation scenarios. For example, under the Multi-Modality Perturbation setting, MUAD achieves an F1-score of 0.9399 with only 4.18% performance degradation, significantly outperforming Eadro, whose F1-score drops to 0.7823 with 13.29% degradation—corresponding to a 9.11 percentage point reduction in relative performance loss. This robustness advantage stems from MUAD’s joint modeling of intra-modal and inter-modal uncertainties. Specifically, the Graph-based Probabilistic Encoder (GPE) captures modality-specific uncertainty by representing features as probabilistic distributions, while the Confidence-aware Fusion Mechanism (CFM) dynamically adjusts fusion weights by down-weighting unreliable modalities and emphasizing more trustworthy ones at the sample level. In contrast, Eadro adopts a fixed fusion strategy that cannot adapt to modality-level reliability variations, resulting in pronounced performance degradation under inter-modal inconsistency.

2) *Random Modality Missing*: In real-world microservice systems, observability data is often incomplete due to collector failures, network partitions, or transient system issues. To evaluate MUAD’s resilience under such conditions, we construct a random modality missing setting, where observability data is partially unavailable at the sample level. Specifically, for each test sample, one or more modalities are randomly masked to simulate unpredictable data loss, with overall missing rates of 20%, 40%, and 60%. The missing pattern is fully randomized across samples, without assuming which modality is absent or

TABLE V
ROBUSTNESS TO RANDOM MODALITY MISSING

Method	Metric	Overall Missing Rate			
		Clean	20%	40%	60%
Eadro	F1	0.9022	0.8362	0.7800	0.7183
	Deg (%)	0.00	7.32	13.54	20.38
MUAD	F1	0.9809	0.9338	0.8888	0.8401
	Deg (%)	0.00	4.80	9.39	14.35

how many modalities are simultaneously unavailable, reflecting the inherent unpredictability of real-world data collection failures.

As shown in Table V, MUAD demonstrates strong robustness across all missing rates. At a 60% overall missing rate, MUAD maintains an F1-score of 0.8401 with 14.35% performance degradation, substantially outperforming Eadro (F1: 0.7183, 20.38% degradation), corresponding to a 6.03 percentage point advantage in relative robustness. Moreover, MUAD exhibits graceful degradation with near-linear performance decay as missing rates increase, whereas Eadro shows a steeper decline, particularly beyond 40% missing. This robustness can be attributed to the Confidence-aware Fusion Mechanism (CFM), which dynamically reallocates fusion weights when modalities are unavailable, allowing MUAD to emphasize remaining reliable modalities. In contrast, Eadro’s fixed fusion strategy cannot adapt to modality absence, resulting in pronounced performance degradation.

In summary, these robustness experiments show that MUAD maintains stable anomaly detection performance under both inter-modal information inconsistency and random modality missing. This demonstrates MUAD’s robustness to degraded and incomplete multimodal observations and highlights the importance of uncertainty modeling and dynamic modality fusion for reliable anomaly detection in practical microservice environments.

E. RQ4: Necessity of Considering High-Confidence Samples in Active Learning

In this section, we construct a variant of MUAD without the High Confidence Samples (w/o HCS) to evaluate the necessity of taking HCS into consideration in the active learning process. The experiment was conducted on Dataset $\mathcal{A}$ which is considered representative as it features a balanced distribution of anomalies and a sufficient amount of high-quality data for reliable evaluation. As shown in the comparative results in Fig. 6, MUAD consistently outperforms the variant without HCS across all query iterations. On Dataset $\mathcal{A}$, as the number of query iterations increases, the F1-score of MUAD with HCS steadily improves from approximately 0.85 to 0.96, whereas the variant without HCS exhibits lower performance with limited improvement. Selecting samples with high $\widehat{\text{TCP}}$ values proves effective in identifying high-confidence instances and assigning

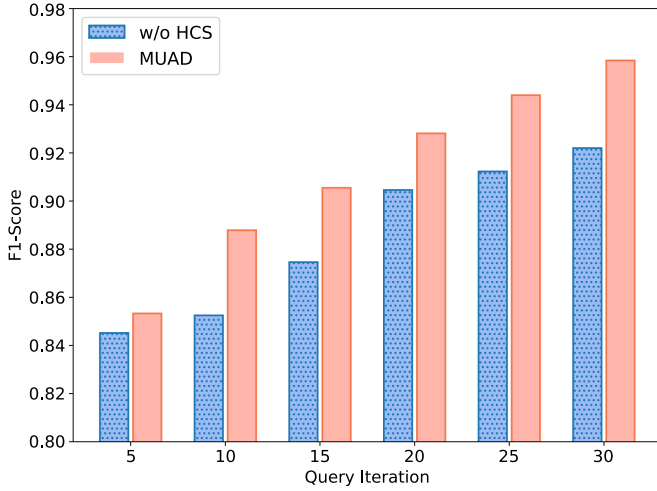


Fig. 6. Impact of high-confidence samples (HCS) selection on active learning performance. The x-axis denotes active learning query iterations, where each iteration corresponds to one querying round followed by model retraining. The orange bars represent MUAD with HCS, while the blue bars show the variant without HCS. Incorporating high-confidence pseudo-labeled samples consistently improves F1-score across query iterations, demonstrating the effectiveness of leveraging reliable predictions to reduce manual annotation costs while maintaining performance.

reliable pseudo-labels, thereby significantly boosting detection performance. Furthermore, the continual inclusion of high-entropy samples into the training set facilitates the discovery of more accurate high-confidence samples over time. Since these high-entropy samples typically correspond to instances where the model is uncertain, their annotation gradually enhances the model's decision confidence and robustness.

In summary, the experimental results confirm that incorporating high-confidence samples is essential for MUAD's active learning strategy, as it effectively reduces annotation costs while maintaining strong detection performance under label scarcity.

F. RQ5: Parameter Sensitivity

In order to balance the ability to model intra- and inter-modal uncertainties, MUAD comprises multiple loss components. We conducted further experiments to analyze the impact of two key loss weights, λ_1 and λ_2 .

Impact of loss weight λ_1 on intra-modal uncertainty modeling: λ_1 controls the weight of the KL divergence term in intra-modal uncertainty modeling. This hyperparameter serves as a regularization factor that balances the trade-off between the compactness and richness of modality-specific information retained in the learned subspace. As shown in Fig. 7, the optimal values of λ_1 for the two datasets are 1.0 and 0.1, respectively. Under these optimal settings, the model tends to predict higher variance for noisy samples and lower variance for high-quality samples.

Impact of loss weight λ_2 on inter-modal uncertainty modeling: λ_2 primarily influences the performance of confidence prediction for each modality. As illustrated in Fig. 7, the optimal value for both datasets is 0.6. By optimizing λ_2 , the model can adjust its final decision strategy according to the dynamically learned weights of each modality.

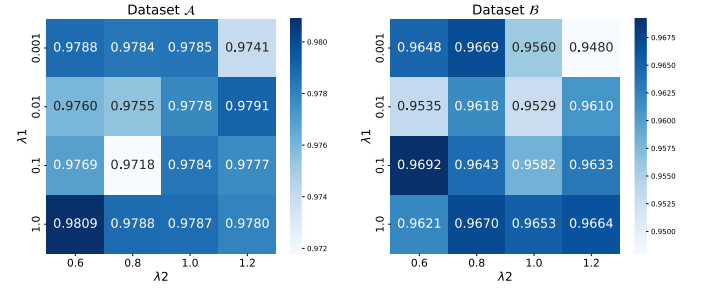


Fig. 7. Sensitivity analysis of loss weights λ_1 and λ_2 on Datasets A and B. The colorbars indicate F1-score performance. Results demonstrate that MUAD maintains consistently high performance across a wide range of hyperparameter values, indicating strong robustness and low sensitivity to parameter selection in practical deployment.

In summary, MUAD demonstrates stable performance across a wide range of loss weight configurations, indicating low sensitivity to precise hyperparameter selection. This robustness can be attributed to the complementary nature of MUAD's loss components: the primary anomaly detection objective ($\mathcal{L}_{\text{ano}}$) dominates the optimization process, while the auxiliary uncertainty modeling terms ($\mathcal{L}_{\text{kl}}$ and $\mathcal{L}_{\text{conf}}$) provide effective regularization as long as their weights fall within reasonable ranges. Such stability is particularly valuable for practical deployment, as it reduces the need for extensive hyperparameter tuning across different microservice environments.

V. RELATED WORK

A. Unimodal Anomaly Detection

Over the years, numerous anomaly detection methods have been proposed to ensure the reliability of large-scale systems. Existing unimodal anomaly detection approaches can be categorized into three types according to the dependent modality: metric-based, log-based, and trace-based methods. Metric-based approaches can be further classified into three categories. Reconstruction-based methods [30], [31], [32] typically reconstruct multivariate time series and detect anomalies based on reconstruction error or reconstruction probability. Prediction-based methods [23], [33], [34] forecast future metrics and trigger anomaly alarms when the actual values deviate from the predicted ones beyond a predefined threshold. Cluster analysis-based approaches [35], [36], [37] embed time series segments into a vector space and apply clustering or distance-based techniques to identify segments that deviate significantly from the majority, thereby flagging them as anomalies.

Logs record critical events during system operation and serve as an important and valuable resource for online anomaly detection. Various log-based anomaly detection techniques [38], [39], [5], [40] have been proposed. For instance, NeuralLog [40] extracts the semantic information of raw log messages using the BERT model and employs a Transformer-based classification model to detect anomalies. In addition, since traces capture the interaction details among microservices, they play an indispensable role in reconstructing the propagation paths of anomalies. Several studies [41], [42], [6] have focused on anomaly detection in trace data. TraceCRL [6] integrates

graph structural information and differentiates between normal and abnormal behavior representations in call chains through a contrastive learning framework. However, as these methods rely solely on a single modality, they are prone to overlooking certain types of anomalies, which limits their ability to ensure the overall security and reliability of the system.

B. Multimodal Anomaly Detection

To overcome the limitations of unimodal anomaly detection, researchers have developed anomaly detection methods based on multimodal data [43], [9], [24], [12], [44], [45], [1], [46], [47], [48]. Specifically, DeepTraLog [43] uses a unified graph to represent the complex structure of traces and the log events embedded within them, and then builds a deep SVDD model based on this to detect anomalies in microservice systems. Anofusion [9] employs GTN to learn the correlations between heterogeneous multimodal data and uses GAT to capture key features, ultimately predicting data patterns for anomaly detection through GRU. AMulSys [44] employs a hierarchical architecture to model traces, metrics, and logs, utilizes graph attention network for trace-metric representation extraction, applies heterogeneous graph Transformer for log representation learning, and introduces category prior guided multimodal data augmentation with multifaceted contrastive learning to address class imbalance in anomaly detection. MicroDig [45] employs a heterogeneous propagation graph to model causal relationships between microservices, combines heterogeneity-oriented random walk algorithm to analyze anomaly propagation patterns, and effectively addresses the root cause localization accuracy issues caused by inconsistency between causality and calling relationships. MULAN [1] proposes a unified framework for multimodal causal structure learning aimed at root cause localization. It leverages a log-tailored language model for log representation learning, employs contrastive learning within a shared latent space to extract both modality-invariant and modality-specific features, and incorporates a KPI-aware attention mechanism to evaluate the reliability of each modality and jointly learn the final causal graph. In comparison with these methods, our approach explicitly accounts for both intra-modal and inter-modal uncertainties, demonstrating superior effectiveness and robustness.

Recently, a few works have leverage the advantages of large language models (LLMs) to improve the performance of AIOPS tasks in microservice systems. For instance, Flow-of-Action [46] proposes an SOP-enhanced multi-agent framework that constrains LLM reasoning through SOP flow mechanisms. By incorporating multiple auxiliary agents, it effectively mitigates the hallucination problems of the ReAct framework in root cause analysis (RCA), thereby significantly improving diagnostic accuracy. LasRCA [47], designed for one-shot scenarios with extremely scarce fault labels, introduces a collaborative framework that integrates LLMs with small classifiers. Through iterative sample selection and LLM-driven feedback annotation, it achieves efficient RCA while balancing accuracy and computational cost. Furthermore, TAMO [48] presents a tool-assisted LLM agent framework that unifies temporally aligned multimodal data representations with domain-specific tools. This design addresses challenges such as restricted context

windows, modal input constraints, and dynamic dependency hallucinations, and demonstrates superior diagnostic performance across heterogeneous datasets and common fault types. Given the growing research interest in combining multimodal data with LLMs for AIOPS in microservices, we plan to incorporate LLMs in future work to achieve more efficient and accurate multimodal anomaly detection.

C. Uncertainty Learning

Uncertainty is primarily categorized into data uncertainty, which is caused by noise within the training data and cannot be mitigated by increasing the amount of data; and model uncertainty, which is mainly due to incomplete training or data and is related to model parameters. Recently, some studies [49], [50], [51], [52] have focused on uncertainty research in deep learning to achieve more robust performance. DUA-Nets [51] dynamically assigns weights to different views of each sample by estimating data uncertainty, thereby effectively integrating multi-view information and mitigating noise effects. DMUE [52] effectively addresses the issue of label ambiguity in facial expression recognition by introducing a multi-branch learning framework to discover the latent distribution within the label space, and leveraging pairwise relationships between instances to estimate uncertainty. Our research focuses primarily on leveraging data uncertainty to improve the performance and robustness of microservice anomaly detection. At the same time, we utilize model uncertainty in active learning to select high-information samples.

VI. CONCLUSION AND FUTURE WORK

In this paper, we proposed MUAD, an uncertainty-aware multimodal anomaly detection framework for microservice systems that explicitly models both intra-modal and inter-modal uncertainties. MUAD integrates a graph-based probabilistic encoder to robustly learn modality-specific representations under noisy and incomplete observations, a confidence-aware fusion mechanism to dynamically adjust modality contributions, and an active learning strategy to reduce labeling cost. Extensive experiments on three benchmark datasets demonstrate that MUAD achieves an average F1-score of 98.10%, outperforming state-of-the-art multimodal methods by up to 7.87%, while consistently exhibiting significantly lower performance degradation under challenging conditions such as inter-modal information inconsistency and random modality missing.

Despite these promising results, we summarize the limitations of our current work and outline corresponding directions for future research: (1) **Log Semantics:** While current log embeddings capture basic patterns, they may miss deep semantic nuances in complex log messages. We plan to incorporate Large Language Models (LLMs) to enhance log semantics and generalization capabilities. (2) **Task Scope:** The current framework of MUAD is primarily optimized for anomaly detection. We intend to extend the uncertainty-aware representations to support a unified framework that simultaneously performs anomaly detection and fine-grained root cause localization for microservice systems.

REFERENCES

- [1] L. Zheng, Z. Chen, J. He, and H. Chen, "Mulán: Multi-modal causal structure learning and root cause analysis for microservice systems," in *Proc. ACM Web Conf. 2024*, 2024, pp. 4107–4116.
- [2] Y. Su, Y. Zhao, C. Niu, R. Liu, W. Sun, and D. Pei, "Robust anomaly detection for multivariate time series through stochastic recurrent neural network," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2019, pp. 2828–2837.
- [3] L. Shen, Z. Li, and J. Kwok, "Timeseries anomaly detection using temporal hierarchical one-class network," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, vol. 33, pp. 13016–13026.
- [4] L. Yang et al., "Semi-supervised log-based anomaly detection via probabilistic label estimation," in *Proc. IEEE/ACM 43rd Int. Conf. Softw. Eng.*, 2021, pp. 1448–1460.
- [5] X. Zhang et al., "Robust log-based anomaly detection on unstable log data," in *Proc. 27th ACM Joint Meeting Eur. Softw. Eng. Conf. Symp. Found. Softw. Eng.*, 2019, pp. 807–817.
- [6] C. Zhang et al., "TraceCRL: Contrastive representation learning for microservice trace analysis," in *Proc. 30th ACM joint Eur. Softw. Eng. Conf. Symp. Found. Softw. Eng.*, 2022, pp. 1221–1232.
- [7] S. Zhang et al., "Efficient and robust trace anomaly detection for large-scale microservice systems," in *Proc. IEEE 34th Int. Symp. Softw. Rel. Eng.*, 2023, pp. 69–79.
- [8] C. Lee, T. Yang, Z. Chen, Y. Su, Y. Yang, and M. R. Lyu, "Heterogeneous anomaly detection for software systems via semi-supervised cross-modal attention," in *Proc. IEEE/ACM 45th Int. Conf. Softw. Eng.*, 2023, pp. 1724–1736.
- [9] C. Zhao et al., "Robust multimodal failure detection for microservice systems," in *Proc. 29th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2023, pp. 5639–5649.
- [10] L. Tao et al., "Giving every modality a voice in microservice failure diagnosis via multimodal adaptive optimization," in *Proc. 39th IEEE/ACM Int. Conf. Automated Softw. Eng.*, 2024, pp. 1107–1119.
- [11] C. Lee, T. Yang, Z. Chen, Y. Su, and M. R. Lyu, "Eadro: An end-to-end troubleshooting framework for microservices on multi-source data," in *Proc. IEEE/ACM 45th Int. Conf. Softw. Eng.*, 2023, pp. 1750–1762.
- [12] J. Huang, Y. Yang, H. Yu, J. Li, and X. Zheng, "Twin graph-based anomaly detection via attentive multi-modal learning for microservice system," in *Proc. 38th IEEE/ACM Int. Conf. Automated Softw. Eng.*, 2023, pp. 66–78.
- [13] Z. Zhao et al., "Chase: A causal heterogeneous graph based framework for root cause analysis in multimodal microservice systems," 2024, *arXiv:2406.19711*.
- [14] N. Zhao et al., "Identifying bad software changes via multimodal anomaly detection for online service systems," in *Proc. 29th ACM Joint Meeting Eur. Softw. Eng. Conf. Symp. Found. Softw. Eng.*, 2021, pp. 527–539.
- [15] Y. Gan et al., "An open-source benchmark suite for microservices and their hardware-software implications for cloud & edge systems," in *Proc. 24th Int. Conf. Architectural Support Program. Lang. Operating Syst.*, 2019, pp. 3–18.
- [16] X. Zhou et al., "Benchmarking microservice systems for software engineering research," in *Proc. 40th Int. Conf. Softw. Eng., Companion Proc.*, 2018, pp. 323–324.
- [17] MUAD, (n.d.). 2025. [Online]. Available: <https://github.com/slg-frank/muad.git>
- [18] A. G. Hawkes, "Markov processes in APL," *ACM SIGAPL APL Quote Quad*, vol. 20, no. 4, pp. 173–185, 1990.
- [19] Y. Wang, Z. Zhu, Q. Fu, Y. Ma, and P. He, "MRCA: Metric-level root cause analysis for microservices via multi-modal data," in *Proc. 39th IEEE/ACM Int. Conf. Automated Softw. Eng.*, 2024, pp. 1057–1068.
- [20] C. Corbière, N. Thome, A. Bar-Hen, M. Cord, and P. Pérez, "Addressing failure prediction by learning model confidence," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, vol. 32, pp. 2902–2913.
- [21] Z. Han, F. Yang, J. Huang, C. Zhang, and J. Yao, "Multimodal dynamics: Dynamical fusion for trustworthy multimodal classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 20707–20717.
- [22] G. Yu, P. Chen, Y. Li, H. Chen, X. Li, and Z. Zheng, "Nezha: Interpretable fine-grained root causes analysis for microservices on multi-modal observability data," in *Proc. 31st ACM Joint Eur. Softw. Eng. Conf. Symp. Found. Softw. Eng.*, 2023, pp. 553–565.
- [23] H. Zhao et al., "Multivariate time-series anomaly detection via graph attention network," in *Proc. IEEE Int. Conf. Data Mining*, 2020, pp. 841–850.
- [24] H. Liu et al., "UAC-AD: Unsupervised adversarial contrastive learning for anomaly detection on multi-modal data in microservice systems," *IEEE Trans. Serv. Comput.*, vol. 17, no. 6, pp. 3887–3900, Nov./Dec. 2024.
- [25] H. Dou, P. Shi, Y. Zhang, P. Chen, and Z. Zheng, "Deanomaly: Anomaly detection for multivariate time series using robust decomposition and memory-augmented diffusion models," *IEEE Trans. Instrum. Meas.*, vol. 74, 2025, Art. no. 3538714.
- [26] J. Yang, Z. Wang, S. Chen, H. He, Y. Hou, and X. Jiang, "HG-PAD: Heterogeneous graph structure learning aided performance anomaly diagnosis in microservice systems," *IEEE Trans. Serv. Comput.*, vol. 18, no. 5, pp. 2916–2931, Sep./Oct. 2025.
- [27] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Representations*, 2015.
- [28] Z. Xie et al., "Unsupervised anomaly detection on microservice traces through graph VAE," in *Proc. ACM Web Conf. 2023*, 2023, pp. 2874–2884.
- [29] S. Xie et al., "TVDiag: A task-oriented and view-invariant failure diagnosis framework for microservice-based systems with multimodal data," *ACM Trans. Softw. Eng. Methodol.*, vol. 35, no. 2, pp. 1–39, 2025, doi: [10.1145/3734868](https://doi.org/10.1145/3734868).
- [30] S. Liu et al., "Time series anomaly detection with adversarial reconstruction networks," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 4, pp. 4293–4306, Apr. 2023.
- [31] Z. Li et al., "Multivariate time series anomaly detection and interpretation using hierarchical inter-metric and temporal embedding," in *Proc. 27th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2021, pp. 3220–3230.
- [32] J. Wang, S. Shao, Y. Bai, J. Deng, and Y. Lin, "Multiscale wavelet graph autoencoder for multivariate time-series anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 72, 2023, Art. no. 2502911.
- [33] A. Deng and B. Hooi, "Graph neural network-based anomaly detection in multivariate time series," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 5, pp. 4027–4035.
- [34] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using LSTMs and nonparametric dynamic thresholding," in *Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2018, pp. 387–395.
- [35] A. Zimek, E. Schubert, and H.-P. Kriegel, "A survey on unsupervised outlier detection in high-dimensional numerical data," *Stat. Anal. Data Mining, ASA Data Sci. J.*, vol. 5, no. 5, pp. 363–387, 2012.
- [36] J. Kim and C. D. Scott, "Robust kernel density estimation," *J. Mach. Learn. Res.*, vol. 13, no. 1, pp. 2529–2565, 2012.
- [37] L. Xiong, B. Póczos, and J. Schneider, "Group anomaly detection using flexible genre models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2011, vol. 24, pp. 1071–1079.
- [38] M. Du, F. Li, G. Zheng, and V. Srikumar, "Deeplog: Anomaly detection and diagnosis from system logs through deep learning," in *Proc. 2017 ACM SIGSAC Conf. Comput. Commun. Secur.*, 2017, pp. 1285–1298.
- [39] W. Meng et al., "LogAnomaly: Unsupervised detection of sequential and quantitative anomalies in unstructured logs," in *Proc. Int. Joint Conf. Artif. Intell.*, 2019, vol. 19, no. 7, pp. 4739–4745.
- [40] V.-H. Le and H. Zhang, "Log-based anomaly detection without log parsing," in *Proc. 36th IEEE/ACM Int. Conf. Automated Softw. Eng.*, 2021, pp. 492–504.
- [41] P. Liu et al., "Unsupervised detection of microservice trace anomalies through service-level deep Bayesian networks," in *Proc. IEEE 31st Int. Symp. Softw. Rel. Eng.*, 2020, pp. 48–58.
- [42] Y. Xu et al., "TraceLingo: Trace representation and learning for performance issue diagnosis in cloud services," in *Proc. 2021 IEEE/ACM Int. Workshop Cloud Intell.*, 2021, pp. 37–40.
- [43] C. Zhang et al., "DeepTraLog: Trace-log combined microservice anomaly detection through graph-based deep learning," in *Proc. 44th Int. Conf. Softw. Eng.*, 2022, pp. 623–634.
- [44] P. Wang, X. Zhang, and Z. Cao, "Anomaly detection for microservice system via augmented multimodal data and hybrid graph representations," *Inf. Fusion*, vol. 118, 2025, Art. no. 103017.
- [45] L. Tao et al., "Diagnosing performance issues for large-scale microservice systems with heterogeneous graph," *IEEE Trans. Serv. Comput.*, vol. 17, no. 5, pp. 2223–2235, Sep./Oct. 2024.
- [46] C. Pei et al., "Flow-of-action: Sop enhanced LLM-based multi-agent system for root cause analysis," in *Proc. Companion ACM Web Conf. 2025*, 2025, pp. 422–431.
- [47] Y. Han, Q. Du, Y. Huang, J. Wu, F. Tian, and C. He, "The potential of one-shot failure root cause analysis: Collaboration of the large language model and small classifier," in *Proc. 39th IEEE/ACM Int. Conf. Automated Softw. Eng.*, 2024, pp. 931–943.
- [48] X. Zhang et al., "Tamo: Fine-grained root cause analysis via tool-assisted LLM agent with multi-modality observation data in cloud-native systems," *IEEE Trans. Serv. Comput.*, vol. 18, no. 6, pp. 4221–4233, Nov. 2025.

- [49] P.-Y. Huang, W.-T. Hsu, C.-Y. Chiu, T.-F. Wu, and M. Sun, "Efficient uncertainty estimation for semantic segmentation in videos," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 520–535.
- [50] F. Kraus and K. Dietmayer, "Uncertainty estimation in one-stage object detection," in *Proc. IEEE Intell. Transp. Syst. Conf.*, 2019, pp. 53–60.
- [51] Y. Geng, Z. Han, C. Zhang, and Q. Hu, "Uncertainty-aware multi-view representation learning," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 9, pp. 7545–7553.
- [52] J. She, Y. Hu, H. Shi, J. Wang, Q. Shen, and T. Mei, "Dive into ambiguity: Latent distribution mining and pairwise uncertainty estimation for facial expression recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 6248–6257.



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